Is ergometer rear elevation a useful perturbation for validating predictive rowing simulations? A pilot EMG study

Roos Duijn[1], Ana de Sousa[2], Achille Roulot[2,3], Guifre Albets[4], Massimo Cenciarini[4]

[1]University of Groningen, University Medical Centre Groningen, the Netherlands. [2]Institut Quimic de Sarria - Universitat Ramon Llull (IQS-URL) Spain, ana.desousa@iqs.url.edu. [3]SeaTech School of Engineering, Université de Toulon,[4]France Universitat Politècnica de Catalunya BarcelonaTech

Introduction and Aim

Predictive musculoskeletal (MSK) simulations could support rehabilitation device design by predicting responses to altered device mechanics [1,2]. Such models should reproduce adaptation to mechanical perturbations, not merely baseline movement. Changing ergometer rear elevation alters the mechanical conditions of rowing and may redistribute lower-limb loading demands [3]. However, it remains unclear whether this perturbation induces consistent changes in muscle recruitment to provide a validation benchmark. Therefore, this pilot study examined (a) whether increasing ergometer rear elevation results in changes in muscle activation duration and muscle activation patterns, and (b) whether this manipulation provides a sufficiently sensitive experimental perturbation for future validation of predictive rowing simulations. We hypothesized that, due to the increased gravitational component associated with greater ergometer inclination, knee, hip, and ankle flexor activity would increase during the early phase of the rowing cycle [3].

Methods

Five healthy participants (20–27 years; 3 males; one with rowing experience) performed rowing trials on a Concept2 ergometer at three inclination angles (0°, 6.3°, and 11.2°, Fig1). Surface EMG signals from seven muscles were synchronized with motion capture data. EMG signals were rectified and smoothed using a 40-ms moving average window. Activation duration was determined using a threshold corresponding to the maximum of 15% of the 95th-percentile EMG amplitude or 3SD above baseline noise. Differences in activation duration were assessed using Friedman tests. Time-normalized muscle activation patterns were compared using one-dimensional Statistical Parametric Mapping (SPM) ($\alpha = 0.05$).

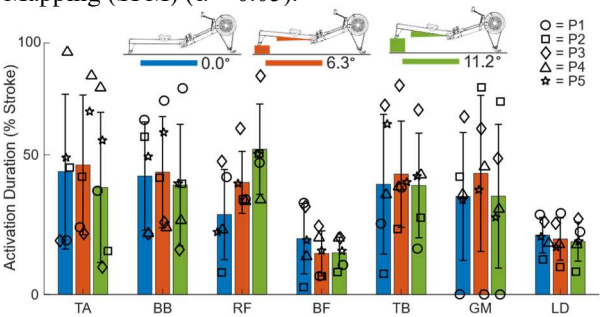


Fig 1. Muscle activation duration (% stroke cycle) for tibialis anterior (TA), biceps brachii (BB), rectus femoris (RF), biceps femoris (BF), triceps brachii (TB), gastrocnemius medialis (GM) and latissimus dorsi (LD). Coloured bar indicate group means and error bars represent ± SD.

Results

Activation duration did not differ significantly between configurations for any muscle. The SPM analysis identified one significant condition effect in the rectus femoris between 8–15% of the rowing cycle (Fig. 2B) However when comparing within subject there was a difference in between inclination angle in muscle activation pattern, showing more RF activation with higher inclination.

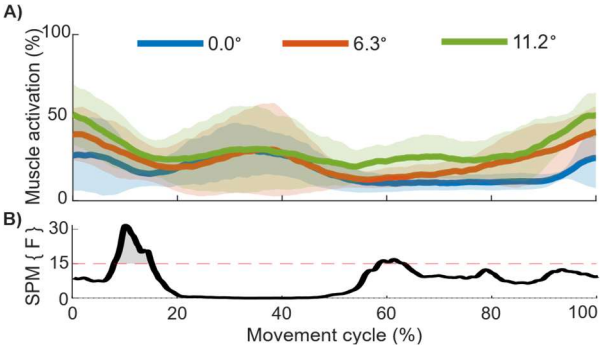


Fig 2.A) Rectus femoris activation patterns during rowing at three ergometer inclination angles. Mean (n = 5) shown as bold lines and SD as shaded areas. B) Repeated-measures SPM ANOVA; the red line indicates the critical threshold, and grey regions indicate significant differences ($p < 0.05$).

Conclusions

Increasing ergometer inclination produced no detectable changes in muscle activation duration and significant differences in the activation profile of only the rectus femoris. These findings suggest that changes in rowing ergometer slide inclination have limited effects at the group level. However, considerable inter-individual variability was observed, indicating that participants adopted different neuromuscular adaptation strategies in response to the altered mechanical conditions. Given the small sample size, the resulting dataset provides an initial experimental basis for future validation of predictive musculoskeletal simulations to support rehabilitation device design.

References

1. Uchida et al, PLoS One, 11:e0163417, 2016.
2. de Sousa et al, IFAC-PapersOnLine, 51:302–307, 2019
3. Hussain et al, Proc 3rd Asia Int Conf Modelling Simul, 242–245, 2009.

Acknowledgements

We thank Pauline Azaïs, Marc Plans, and Xu for their assistance with the experiments. This work was supported by an Early Career Stimulus Award from the Health in Context Research Institute.